

BAP: A Blockchain-Assisted Privacy-Preserving Authentication Protocol With User-Controlled Data Linkability for VANETs

Zijian Bao , Debiao He , *Member, IEEE*, Huaqun Wang , Min Luo , and Cong Peng 

Abstract—With the proliferation of vehicular networking and data in the era of Vehicular Ad-hoc Networks (VANETs), ensuring privacy-preserving authentication and data privacy during analysis has emerged as a pivotal research focus. In this article, we propose a blockchain-assisted privacy-preserving authentication protocol BAP with user-controlled data unlinkability for VANETs. We leverage Pointcheval-Sanders (PS) signatures to design a privacy-preserving authentication protocol that supports user traceability and revocation of malicious users. Additionally, we introduce an auxiliary data processor (DP) in our model to analyze VANET data with explicit and implicit linkability. The DP is capable of linking messages on the same or different topics, enabling minimal privacy disclosure during Big Data analysis. We provide a detailed proof and demonstrate that our scheme satisfies the required properties. Finally, performance evaluations demonstrate the efficiency of our proposed scheme. While supporting the mentioned functionalities, the gas cost is limited to a small range, and the signature size is only 354 bytes.

Index Terms—Blockchain, Authentication, Privacy-Preserving, Data Linkability.

I. INTRODUCTION

THE Internet of Vehicles (IoV) [1], [2] is a network that leverages Big Data to facilitate information exchange among “people-vehicles-road-cloud” through specific communication protocols and standardized data interactions. It seamlessly connects vehicles, road infrastructure, cloud computing, and individuals, enabling cooperative and intelligent traffic management. As shown in Fig. 1, the common VANETs [3], [4] contain vehicles equipped with on-board units (OBU), roadside units (RSUs), Internet and the trusted authority (TA). There exists three types of communication: vehicle-to-vehicle (V2V), and vehicle-to-RSU (V2R), wired/wireless Internet. The OBUs facilitate seamless data exchange among vehicles, while RSUs act as communication gateways, enhancing network coverage and enabling efficient connectivity between vehicles and the Internet. The TA plays a crucial role in ensuring the security and integrity of the VANET by managing authentication, authorization, and cryptographic key distribution, fostering trust among the participating entities.

In spite of the ability of the VANET architecture mentioned above to facilitate information exchange and data convergence, the current communication framework is exposed to various security vulnerabilities [5]. Particularly, when messages are transmitted wirelessly between vehicles, they become susceptible to eavesdropping, tampering, and other malicious activities. The compromise of critical vehicle-related information, including speed, location, and personal data, can have severe implications and potentially result in disastrous outcomes. Therefore, it is imperative to address these vulnerabilities and safeguard the integrity and confidentiality of data exchanged within VANETs.

Ensuring secure and trustworthy communication among Vehicle-to-Vehicle (V2V) and Vehicle-to-Roadside (V2R) connections is a pivotal research focus [6]. Authentication mechanisms play a vital role in verifying the identity of vehicles and Roadside Units (RSUs), facilitating secure communication, and safeguarding against malicious entities. The implementation of robust authentication mechanisms is essential to foster trust among the participating entities and to mitigate the risks associated with unauthorized access and information compromise in the VANET environment.

However, the authentication schemes still confront the issue of privacy leakage, which has the potential to disclose vehicle identity information. Several privacy-preserving authentication

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protocols [4], [7], [8] for VANETs have already been proposed, yet most do not adequately consider the traceability of malicious users and the implementation of efficient revocation mechanisms. Furthermore, in the era of Big Data, it is crucial to incorporate vehicle data analysis into our model. The challenge lies in achieving the minimum level of privacy leakage granularity while simultaneously maximizing the utilization of data.

In this article, inspired from ideas for data protection in group signatures [9], [10], we introduce a data processor for the analysis of data in the context of vehicular networks. To facilitate effective analysis of relevant VANETs data, we propose the use of a data label, referred to as “*topic*”, along with two types of data linkability: implicit linkability and explicit linkability. On the one hand, implicit linkability allows the data processor to establish connections between data within the same topic for a particular vehicle. This enables the aggregation and analysis of related data, providing insights into vehicle behavior and performance within specific contexts. On the other hand, explicit linkability comes into play when dealing with data from different topics for the same vehicle. In such cases, the data processor can establish connections only when explicit proof of linkage is provided by the vehicle actively. This ensures that data from different topics can be linked together for comprehensive analysis, while maintaining data privacy and integrity. Our approach aims to strike a balance between data analysis and privacy protection. Messages from the same vehicle on the same topic can be directly analyzed by the data processor, without requiring explicit declaration. However, for messages from the same vehicle on different topics, proactive declaration of linkages by the vehicle is necessary to enable proper analysis. We call them user-controlled linkability, i.e., the user is in control of the processor’s links to this data.

Further, blockchain [11] and smart contracts [12] have emerged as transformative technologies, revolutionizing industries across the globe. Blockchain, a decentralized and immutable ledger, ensures transparent and secure record-keeping by cryptographically linking individual transactions into an unalterable chain. Complementing this, smart contracts are self-executing agreements with predefined conditions, automatically executing transactions without intermediaries. Leveraging these innovations, we design two smart contracts to store the key information. This contract harnesses the transparency of blockchain to maintain data integrity and facilitate efficient management of registered vehicles by authorized entities.

A. Our Contributions

- 1) First, we introduce BAP, a privacy-preserving authentication protocol enhanced by blockchain technology. Our protocol leverages PS group signatures, non-interactive zero-knowledge proofs (ZKPOK) and signature of knowledge (SOK) techniques to ensure privacy protection. Additionally, we employ accumulators to facilitate the registration of legitimate user identities and the revocation of malicious identities. The utilization of blockchains enables us to record critical data securely, mitigating the risk of data tampering by malicious participants.

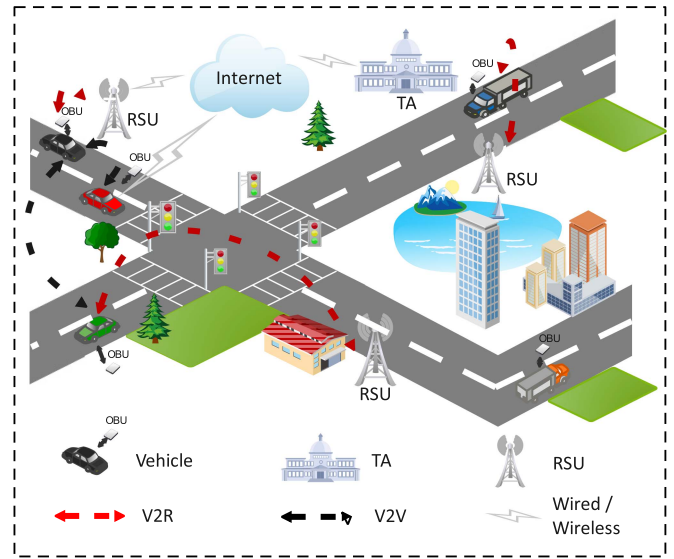


Fig. 1. Architecture of VANETs.

- 2) Second, we present a comprehensive security analysis of our proposed protocols, namely ZKPOK and SOK. Additionally, we conduct an analysis of the properties of BAP, in terms of supporting message authentication, privacy preservation, conditional traceability, unlinkability, resistance against common attacks and user-controlled linkability.
- 3) Third, we provide a detailed performance evaluation, focusing on three key aspects: blockchain cost, authentication cost, and simulation cost. For our blockchain test network, we employ Ethereum as the underlying platform. To simulate vehicle networking, we utilize both VanetMobiSim and NS-2.

B. Organization

In Section II, we review the related work. In Section III, we present the system model and outline the specific security requirements. In Section IV, we introduce the notations and preliminary, and propose two designed smart contracts, further give the high-level description. In Section V, we present our proposed privacy-preserving authentication protocol with user-controlled data linkability. In Section VI, we conduct a thorough security analysis of our proposed protocol. In Section VII, we analyze the performance of the solution from three aspects. In Section VIII, we conclude the article.

II. RELATED WORK

In this section, we review the related authentication protocols for VANETs. Raya and Hubaux [13] proposed a road condition information transferring system with a Public Key Infrastructure (PKI) system. However, it requires large number of public-private key pairs and user certificates, which reduces the efficiency of system usage. After that, Lu et al. [14] made use of anonymous certificates to conduct a privacy-preserving authentication protocol. The vehicle can obtain a temporary

anonymous certificate when it passes by a RSU. However, frequent communication between the vehicle and RSU may make the system inefficient. Wasef et al. [15] presented an expedite message authentication protocol. They adopted the PKI and Certificate Revocation Lists (CRLs) and used the keyed hash message authentication code to achieve revocation check process. Soleymani et al. [16] proposed a security model named AFPM for 5G-VANET, utilizing an authentication mechanism with cuckoo filter for illegitimate node detection and a plausibility model based on fuzzy logic to handle inaccurate information and misbehavior nodes. Wen et al. [17] presented a novel message physical layer assisted message authentication (PAA) framework in vehicular communication networks, utilizing temporal and spatial uniqueness in physical layer channel responses to establish trust between vehicles, achieving high efficiency, minimal authentication delay, and strong security without compromising the requirements. Wang et al. [8] proposed a two-factor lightweight privacy-preserving authentication by using message-authentication-code (MAC) and hash operations. However, it requires the CA to maintain a large table of vehicle messages and system keys. However, the existing schemes both have the weakness that a reliable third party necessitates significant storage capacity to accommodate all the certificates.

Some researchers have turned to identity-based cryptosystems to avoid mass storage of certificates. Zhang et al. [18] gave an identity-based batch signature verification scheme for vehicular communications. It utilized the group testing technique, enabling the detection of invalid signatures through a minimal number of batch verifications. Chim et al. [19] proposed a software-based solution by using two shared secrets to enhance the security of [18]. However, it still suffers from the impersonation attack presented by Horng et al. [20]. To fix the attack, He et al. [21] put forward an efficient identity-based conditional privacy-preserving authentication scheme for VANETs. They used a schnorr-like mechanism to conduct the protocol and assumes that the vehicle is equipped a tamper-proof OBU.

With the popularity of blockchain technology, research scholars are considering the use of blockchain to secure critical data and design authentication protocols accordingly. Lu et al. [22] used a Merkle Patricia tree to design a distributed authentication protocol for VANETs. However, it requires a multi-interaction between a vehicle and the trusted party. Gabay et al. [23] used a token-based mechanism and a Pederson commitment scheme to realize anonymous authentication. Feng et al. [7] presented a blockchain-assisted privacy-preserving authentication system BPAS without an online registration center. They used fuzzy extractor and attribute-based encryption to preserve vehicle privacy. Li et al. [24] introduced a new accountable attribute-based authentication with fine-grained access control, achieving fine-grained access control, anonymity, and public accountability in user authentication. They also presented the first attribute-based, fair, anonymous, and publicly traceable crowdsourcing scheme on blockchain, balancing anonymity and accountability while ensuring fairness in worker competition. Lin et al. [4] combined the blockchain technology and a key derivation algorithm BCPA to achieve a certificate management, which meets both

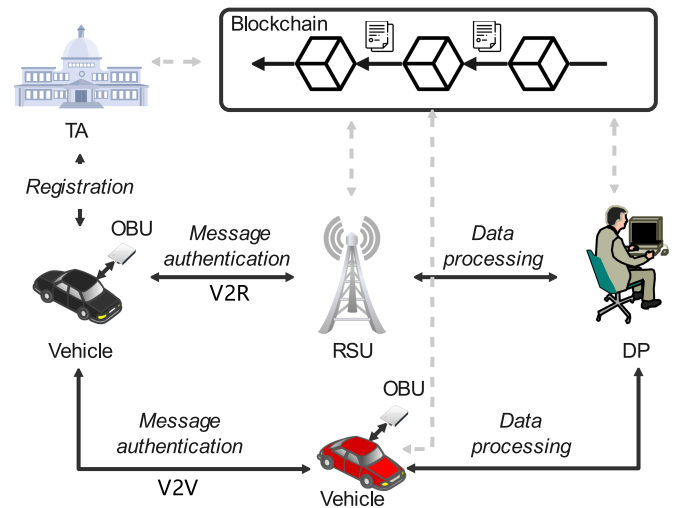


Fig. 2. System model.

anonymity and traceability. However, none of them consider the data linkability after the authentication.

III. PROBLEM OVERVIEW

In this section, we present the system model and security requirements.

A. System Model

VANET is a wireless sensor network based on vehicles communicating with each other, which can provide services such as intelligent traffic management, vehicle security, driver assistance, etc. Our system model includes (see Fig. 2) the following five entities: TA (Trust Authority), vehicle, RSUs (Roadside Unit), blockchains and DPs (data processors).

- **Trust Authority (TA):** It is responsible for ensuring secure communication between vehicles and RSUs. TA protects communication between vehicles and RSUs by providing services such as security certificates and key management to prevent malicious attacks and data leakage. It has sufficient computation and communication capabilities and is considered trusted. In addition, TA is also involved in maintaining the blockchain system to protect data from tampering. Moreover, if necessary, TA can trace a vehicle's real identity and revoke it.
- **Vehicle:** It is equipped with OBU (On-Board Unit), which is used to receive and send vehicle information. OBU can get information from RSU and other vehicles, such as traffic congestion, accident alerts, etc., and also can send its own information to other vehicles and RSU.
- **Roadside Unit (RSU):** It is installed on the road, acts as an access point between vehicles and the network (including blockchain). RSU can provide information on vehicle location, traffic flow, etc., and can also send road condition information, traffic management instructions to vehicles. It connects to nearby vehicles and verifies the validity of received messages and uploads data to DPs for instructions.

- **Blockchain:** It is a decentralized and distributed database technology. Blockchain technology can be used to manage and authenticate the identity and information of vehicles. This can help avoid fraud and unnecessary disputes, and increase the management and control of vehicles. Specifically, the blockchain supports two types of smart contracts, i.e., authentication contract to store vehicle registration information and support vehicle revocation; and transaction contract to store data authentication information.
- **Data Processor (DP):** DP can process data from V2V and V2R to provide more efficient instructions for vehicles and better guide on the work of IoV. DP can implicitly link data from the same scope for the same vehicle. While for data from different scopes for the same vehicle, DP can only link data after the vehicle actively provides explicit link proof. DP can analyze the data of the same vehicle in the same scope, and in different scopes (if explicit link proof is provided).

B. Security Requirements

To meet the necessary security issues in real-world scenarios, a feasible blockchain-assisted privacy-preserving authentication protocol for VANETs should fulfill the following vital security requirements.

- **Message authentication:** The vehicle and RSU have the capability to authenticate the received message (including integrity and identity legality). Additionally, the systems are designed to detect any potential attempts at message manipulation. Note we consider the V2V and V2R communications, while other communication is secured by the https protocol, which is out of our goals.
- **Privacy Preservation:** This property requires that the vehicle's identity can be hidden. Only when necessary, the TA can open the vehicle's identity. Therefore, the scheme can effectively protect vehicle's privacy.
- **Conditional Traceability:** This property requires that only TA can open the vehicle's identity when there is malicious behavior occurring. Therefore, the scheme can effectively protect vehicle privacy and provide traceability. Therefore, TA can take corresponding measures to revoke the identity of malicious vehicles and perform regulatory actions (e.g., law enforcement, penalty).
- **Unlinkability:** In order to avoid detection by potential adversaries, it is difficult to link two messages sent by the same vehicle based on previous communication history.
- **Resistance against other common attacks:** The scheme should resist against commonly encountered attacks in VANETs, such as replay attacks, modification attacks, and impersonation attacks.

In addition, we also emphasize two additional requirements.

- **Implicit linkability:** The DP can obtain the message/SOK pair accompanied with a pseudonym, generated by the vehicle for a particular scope scp (e.g., parking payment, motorway toll). For the same scope, the vehicle's pseudonym is the same. While for the different scopes, pseudonymous

TABLE I
USED NOTATIONS

Notations	Descriptions
λ	The security parameter
pp	The public parameters
$e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$	A type-3 bilinear map
p	The order of $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$
$\mathcal{H}_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$	a hash function
$\mathcal{H}_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$	a hash function
(x, y_1, y_2)	the group private key gmsk
$(\tilde{g}, \tilde{X}, \tilde{Y}_1, \tilde{Y}_2)$	the group public key gmpk
ask	the private key for the accumulator
$\text{apk} = \tilde{h}^{\text{ask}}$	the public key for the accumulator
dsk	the private key of the DP
$\text{dpk} = \tilde{h}^{\text{dsk}}$	the public key of the DP
(sk_i, η)	the vehicle i 's private group key gsk_i
uid	a unique identification
tag	a bit denoting "granted" or "revoked"
Acc	the accumulated value
ZKPOK	Zero-knowledge proof of knowledge
SOK	Signature of knowledge
m	the message
π	the group signature
scp	a scope that identifies a use case
cnym	an encrypted pseudonym
$\text{nym} = \mathcal{H}_1(\text{scp})^{\text{sk}_i}$	a pseudonym
txid	a transaction id
Ω	an explicit link proof

signatures for different scopes (even from the same vehicle) cannot be linked. The property is straight as DP can analyze the data of the same vehicle in the same scope.

- **Explicit linkability:** After the signatures has been generated, the vehicle can actively claimed and linked them, i.e., the vehicle can generates a proof to prove that two signatures for different scopes belong to the same vehicle. This property is also necessary, e.g., sometimes vehicle management agencies need to calculate the vehicle's parking payment, motorway toll and other spending, and determine whether a certain amount is reached, to make an overall discount.

IV. OUR CONSTRUCTION

A. Notations

For convenience, the notations used in our paper are listed in Table I.

B. Preliminary

1) Bilinear Pairing: Let $\mathbb{G}_1, \mathbb{G}_2$, and $\mathbb{G}_T$ be cyclic groups with the order p . We say that a map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a bilinear map if it satisfy the following properties [25]. Note

that we employ a type-3 bilinear pairing, which implies that $\mathbb{G}_1$ and $\mathbb{G}_2$ are distinct groups and there exists no homomorphism between them that can be efficiently computed.

- **Bilinear:** For $\forall a, b \in \mathbb{Z}_p^*$ and $\forall g \in \mathbb{G}_1, \forall \tilde{g} \in \mathbb{G}_2$, $e(g^a, \tilde{g}^b) = e(g, \tilde{g})^{ab}$.
- **Non-degenerate:** For $\forall g \in \mathbb{G}_1, \tilde{g} \in \mathbb{G}_2$, $e(g, \tilde{g}) \neq 1$, where 1 denotes the identity element in $\mathbb{G}_T$.
- **Computability:** For $\forall g \in \mathbb{G}_1, \tilde{g} \in \mathbb{G}_2$, $e(g, \tilde{g})$ can be efficiently computable.

2) **ZKPOK:** Zero-knowledge proof of knowledge (ZKPOK) [26] is a special kind of zero-knowledge proof that allows a prover to convince a verifier that he knows a witness w satisfying a relation $\mathcal{R}$ with a public statement x without revealing the witness itself. We make use of the notation $\text{ZKPOK}\{(w) : (w, x) \in \mathcal{R}\}$ for representing a ZKPOK proof. Currently, Σ -protocol [27] is a well-used three-move ZKPOK protocol. It can be further transformed to a Signature of Knowledge (SOK) [28] with Fiat-Shamir heuristic. Compared to ZKPOK, it can additionally be seen as a signature for the message m . It consists of three probabilistic polynomial time (PPT) algorithms, which should meet the properties of completeness, soundness, zero-knowledge.

- **Setup(λ):** Given the security parameter λ , this algorithm outputs public parameters pp .
- **GenProof($\text{pp}, m, w, x, \mathcal{R}$):** Given the public parameters pp , a message m , a witness/statement pair $(w, x) \in \mathcal{R}$, this algorithm outputs a proof π . We use the equation $\text{SOK}\{(w) : (w, x) \in \mathcal{R}\}(m)$ to represent the process.
- **VrfProof(pp, m, x, π):** Given the public parameters pp , a message m , a statement x and a proof π , this algorithm checks the validity of π . If valid, this algorithm outputs 1; otherwise, outputs 0.

3) **PS Signature:** Pointcheval-Sanders (PS) [29] signature is a privacy-perserving signature scheme that employs type-3 bilinear pairing to achieve shorter signature size. The signature consists of only two elements, while the digital signature technique ensures the authenticity and integrity of the signed data. Moreover, the signature σ can be randomized to a new randomized signature σ' , while keeping the validity of the signature. Overall, this scheme presents a promising solution for practical applications that require privacy protection. The PS Signature Φ consists of four PPT algorithms.

- **Setup(λ):** Given the security parameter λ , this algorithm outputs public parameters $\text{pp} = \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, p\}$, where $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a type-3 bilinear pairing, p is the order of $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$.
- **KenGen(pp):** Given the public parameters pp , it randomly chooses $\tilde{g} \in \mathbb{G}_2$, and computes $x, y \in \mathbb{Z}_p^*$, sets $(\tilde{X}, \tilde{Y}) \leftarrow (\tilde{g}^x, \tilde{g}^y)$, this algorithm outputs the key pair (sk, pk) , $sk = (x, y)$, $pk = (\tilde{g}, \tilde{g}^x, \tilde{g}^y)$.
- **Sign(m, sk):** Given a message m , it randomly chooses $g \in \mathbb{G}_1^*$, this algorithm outputs a signature $\sigma \leftarrow (\sigma_1, \sigma_2) \leftarrow (g, g^{(x+y \cdot m)})$.
- **Verify(m, pk, σ):** Given a message m , a public key pk and a signature σ , it parses σ as (σ_1, σ_2) , then checks $\sigma_1 \neq 1_{\mathbb{G}_1}$ and $e(\sigma_1, \tilde{X} \cdot \tilde{Y}^m) = e(\sigma_2, \tilde{g})$. If passed, this algorithm outputs 1; otherwise, outputs 0.

4) **Group Signature Based on PS Signature:** The group signature [29] based on PS signature Ψ consists of six PPT algorithms.

- **GSetup(λ):** Given the security parameter λ , it sets public parameters $\text{pp} = \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, p\}$. It randomly chooses $\tilde{g} \in \mathbb{G}_2$, $x, y \in \mathbb{Z}_p^*$, computes $(\tilde{X}, \tilde{Y}) \leftarrow (\tilde{g}^x, \tilde{g}^y)$, this algorithm outputs the public parameters pp , the group private key $\text{gsk} = (x, y)$ along with a generator $g \in \mathbb{G}_1$, the group public key $\text{gmpk} = (\tilde{g}, \tilde{g}^x, \tilde{g}^y)$.
- **PKIJoin(i, λ):** The user i uses $\Phi.\text{Setup}(\lambda)$ to get pp and $\Phi.\text{KenGen}(\text{pp})$ to generate a public/private key pair $(\text{usk}_i, \text{upk}_i)$ and sends upk_i to the certification authority.
- **GJoin:** The user i who wants to join the group, first initiates an interactive protocol with the group manager (abbreviated as GM). The user randomly chooses $\text{sk}_i \leftarrow \mathbb{Z}_p^*$, sets $(\gamma, \tilde{\gamma}) \leftarrow (g^{\text{sk}_i}, \tilde{Y}^{\text{sk}_i})$, computes $\sigma \leftarrow \Phi.\text{Sign}(\gamma, \text{usk}_i)$ and sends σ to the GM. Then, the GM checks the validity of σ and checks whether $e(\gamma, \tilde{Y}) = e(g, \tilde{\gamma})$. Then, the user proves that he knows the secret value sk_i , i.e., sends $\text{POK}\{(\text{sk}_i) : \gamma = g^{\text{sk}_i} \wedge \tilde{\gamma} = \tilde{Y}^{\text{sk}_i}\}$. Then, the GM randomly chooses $t \leftarrow \mathbb{Z}_p^*$ and computes $\eta \leftarrow (\eta_1, \eta_2) \leftarrow (g^t, (g^x \cdot \gamma^y)^t)$ which is a signature of sk_i (i.e., $\eta = (g^t, g^{t(x+y \cdot \text{sk}_i)})$). At last, the GM stores the entry $(i, \gamma, \tilde{\gamma}, \sigma)$ in his own register, and sends η to the user. The user i sets his private group key gsk_i as (sk_i, η) .
- **GSign(m, gsk_i):** The user randomly chooses $r \in \mathbb{Z}_p^*$ and computes a re-randomized signature $(\eta'_1, \eta'_2) \leftarrow (\eta_1^r, \eta_2^r)$, then computes a signature of proof $\text{SOK}\{(\text{sk}_i) : e(\eta'_1, \tilde{X} \cdot \tilde{Y}^{\text{sk}_i}) = e(\eta'_2, \tilde{g})\}(m)$ as a group signature π on m .
- **GVerify(m, gpk_i, π):** The verifier checks whether π is a valid SOK. If valid, the verifier outputs 1; otherwise, outputs 0.
- **GOpen(m, gmsk, μ):** When we need to open a user's signature, the GM searches all entries $(i, \gamma_i, \tilde{\gamma}_i, \sigma_i)$ in his own register whether $e(\eta_2, \tilde{g}) \cdot e(\eta_1, \tilde{X})^{-1} = e(\eta_1, \tilde{\gamma}_i)$ holds. Until he gets a match, he outputs the corresponding entry (i, γ_i, σ_i) where i is the user's identity and outputs a proof of knowledge of a valid $\tilde{\gamma}_i$. Anyone can check the validity of the proof.

5) **Accumulator:** We make use of an accumulator [30], [31] with one-way domain. An accumulator can combine a set of value $\{x_i\}_{i \in [n]}$ into an accumulated value acc . For a x_i , we have a witness w_i , which can be used for proving that x_i is indeed accumulated into acc . If values can be dynamically added to or removed from an accumulator, it is a dynamic accumulator. The accumulator consists of four PPT algorithms.

- **Setup(λ):** Given the security parameter λ , this algorithm randomly chooses $sk \in \mathbb{Z}_p^*$, $\tilde{h} \in \mathbb{G}_2$, computes $pk = \tilde{h}^{sk}$ and outputs a public/private key pair (pk, sk) .
- **Eval($\{x_i\}_{i \in [n]}, sk$):** Given a set of value $\{x_i\}_{i \in [n]} \in \mathbb{Z}_p^*$, this algorithm outputs an accumulated value $\text{acc} = \tilde{h}^{\sum_{i=1}^n (x_i + sk)}$.
- **WitnessGen(acc, x_j):** Given an accumulated value acc and a value x_j , this algorithm outputs a witness $w = \tilde{h}^{\sum_{i=1, i \neq j}^n (x_i + sk)}$.

- $\text{Verify}(acc, w, x_j)$: Given an accumulated value acc , a witness w and a value x_j , this algorithm checks whether $acc = w^{x_j}$. If valid, the verifier outputs 1; otherwise, outputs 0.

6) *Blockchain and Smart Contract*: Blockchain [11] employs cryptographic techniques to provide data security and tamper resistance. It serves as a decentralized distributed database system. Blockchain is made up of a number of chronologically ordered blocks, each of which includes the hash value of preceding blocks as well as the current newly added transactions. Through a decentralized trust mechanism and transparent transaction records, blockchain increases the reliability and efficiency of transactions while lowering costs and risks. Recently, the smart contracts [12] on the blockchain have received a lot of attention. Smart contracts are automated contracts that can execute, validate, and enforce the terms of a contract digitally without the assistance of a third party. Blockchain technology is used in smart contracts to guarantee the validity and tamper resistance character of the contract as well as the capacity to automatically enforce the conditions. By automating the execution of contracts, it lowers transaction costs, saves time, and boosts transaction reliability.

C. Design of Smart Contract

We design two smart contracts to store the key information. The first contract authentication contract is used for store the vehicle registration information and support vehicle revocation. Namely, it stores a entry (uid, tag, Acc) , where uid is a randomly selected number but unique to each vehicle, tag is a bit (1 denotes the member was granted; 0 denotes the member was revoked), Acc is the accumulated value after granting or revoking the member. It is depolyed by the the TA. It consists of four algorithms `UserList`, `updateUser`, `getAcc` (see Algorithm 1). By providing the smart contract address and the necessary parameters (such as the algorithm name and arguments), one can invoke the algorithms. The `UserList` is a constructor algorithm, which is only executed once when a contract is created. The `updateUser` algorithm can only be invoked by the contract owner (i.e., TA), it is used for granting or revoking a vehicle. The `getAcc` algorithm provides the latest accumulated value to the invoker, which will be later used to prove that the vehicle's unique identification uid has indeed been accumulated to the latest accumulation value, i.e., to show that the vehicle's identity is legitimate.

Another contract is the transaction contract to store data authentication information. It mainly stores two parts, the first part is a group signature π (i.e., a SOK proof in our paper), the group signature is used for providing authentication for the V2R/V2V messages; the second part is a vehicle's encrypted pseudonym $cnym$, which contains $(cnym1, cnym2)$. The encrypted pseudonym will be later decrypted into a pseudonym, which is used for providing implicit/explicit linkability. It is depolyed by the the TA. It consists of three algorithms `TxList`, `addNym`, `getNym` (see Algorithm 2). The `TxList` is also a constructor algorithm. The `addTx` is used for adding the group signature and encrypted pseudonym. The `getTx` is used for obtaining the corresponding authentication information.

Algorithm 1: Authentication Contract.

Require: Function name, invoked parameters

Ensure: Setting up functions:

address owner; % The address of owner is TA.

Struct `UserList` {

uint256 uid; % the user's unique identification.

bool tag; % the user's authorization status.

uint256 [2] acc; % the accumulated value.

}

`UserList[]` private userList;

constructor `UserList` public { % Constructor, automatically executed during the deployment.

owner = msg.sender; % Initialize the issuer as manager.

}

function `updateUser(uid, tag, acc)` {

require(msg.sender==owner);

userList.push(`UserList`(uid, tag, acc));

}

function `getAcc()` view {

`UserList` user = userList[arclist.length - 1];

return user.acc;

}

Algorithm 2: Transaction Contract.

Require: Function name, invoked parameters

Ensure: Setting up functions:

address owner; % The address of owner is TA.

Struct `TxList` {

uint256 [9] gs; % a group signature.

uint256 cnym1; % the first part of a pseudonym.

uint256 cnym2; % the second part of a pseudonym.

}

mapping (uint256 => `TxList`) txlist;

constructor `TxList` public { % Constructor, automatically executed during the deployment.

owner = msg.sender; % Initialize the issuer as manager.

}

function `addTx(txid, gs, cnym1, cnym2)` {

txlist[txid].gs = gs;

txlist[txid].cnym1 = cnym1;

txlist[txid].cnym2 = cnym2;

}

function `getTx(txid)` view {

return (txlist[txid].gs, txlist[txid]

.cnym1, txlist[txid].cnym2);

}

D. High-Level Description

- Step 1 System Initialization: Given the security parameter λ , this step initializes the public parameters and blockchain, and deploys the smart contracts

authentication contract and transaction contract on the blockchain.

- **Step 2 Registration:** Given the public parameters pp , the vehicle i registers with TA and obtains a signature η on sk_i as his credential, the TA uploads a vehicle registration information to the authentication contract.
- **Step 3 Message Signing:** Given a message m , the vehicle signs the message m to obtain a group signature π (i.e., a SOK proof), which ensures that it has a valid re-randomized signature η' of η , along with a encrypted pseudonym $cnym \leftarrow (cnym_1, cnym_2)$, where $cnym$ is a ciphertext of a pseudonym $nym = \mathcal{H}_1(scp)^{sk_i}$, where scp is a scope indicating an application scenario (e.g., parking payment). The nym is further used for providing implicit/explicit linkability in Step 5.
- **Step 4 Message Verification:** Given a message m , a scope scp , a signature π and a encrypted pseudonym $cnym$, the vehicle/RSU must validate the validity of π . If π is determined to be valid, the vehicle/RSU will upload the data authentication information $(\pi, cnym)$ to the authentication contract for further processing by the data processor (DP).
- **Step 5 Message Link:** Given some message/signature/encrypted pseudonym pairs $\{m_i, \pi_i, cnym_i\}_{i \in [n]}$, this step provides two types of linkability: *implicit* or *explicit*. The DP first decrypts the encrypted pseudonym $cnym_i$ to obtain the pseudonym nym_i , where $nym_i = \mathcal{H}_1(scp)^{sk_i}$. For implicit linkability, if $nym_i = nym_j$, it indicates that these two messages belong to the same vehicle for the same scope. For explicit linkability, the vehicle must provide an explicit link proof for a list of messages and send the proof to the DP. Then, the DP can verify the proof and determine that these messages belong to the same vehicle.
- **Step 6 Tracking and Revocation:** Given a group signature π on a message m , the TA is able to open the signature and extract the identity i of the vehicle. Moreover, the TA can issue a revocation instruction to the authentication contract to revoke the vehicle. This enables the system to track the behavior of the vehicle and prevent it from accessing the system in case of malicious activity or misbehavior.

V. PROPOSED SCHEME

In this section, we present the detailed construction of the scheme. The overview of BAP is shown in Fig. 3.

A. System Initialization

This step initializes the public parameters and blockchain. The TA needs to deploy two smart contracts.

- **Basic initialization.** The TA chooses a type-3 bilinear pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, p is the order of $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$. The TA randomly chooses $g, h \in \mathbb{G}_1$, $\tilde{g}, \tilde{h} \in \mathbb{G}_2$, $x, y_1, y_2 \in \mathbb{Z}_p^*$, computes $(\tilde{X}, \tilde{Y}_1, \tilde{Y}_2) \leftarrow (\tilde{g}^x, \tilde{g}^{y_1}, \tilde{g}^{y_2})$. The TA also chooses two hash functions: $\mathcal{H}_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$ and $\mathcal{H}_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$. The TA stores (x, y_1, y_2) as

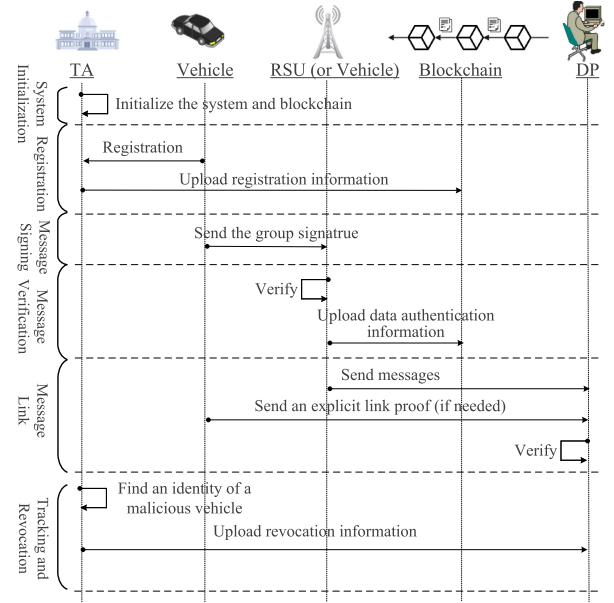


Fig. 3. Overview of BAP.

the group private key gsk . The $(\tilde{g}, \tilde{X}, \tilde{Y}_1, \tilde{Y}_2)$ is the group public key $gmpk$. The TA also randomly chooses $ask \in \mathbb{Z}_p^*$, computes $apk = \tilde{h}^{ask}$, and sets (ask, apk) as the key pair for an accumulator. The initialized accumulated value is set to h .

- The DP randomly chooses $dsk \in \mathbb{Z}_p^*$, computes $dpk = \tilde{h}^{dsk}$, sets (dsk, dpk) as the key pair used for encryption. The DP sends dpk to the TA. The TA outputs the public parameters $pp = \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, p, g, h, \tilde{g}, \tilde{h}, \mathcal{H}_1, \mathcal{H}_2, \tilde{X}, \tilde{Y}_1, \tilde{Y}_2, apk, dpk\}$.
- **Blockchain initialization.** The TA sets the initialization parameters of a blockchain and maintains the blockchain together with other nodes (possibly multiple nodes within the TA or other trusted institutions outside the TA). For simplicity, TAs can also join an existing blockchain system directly.
- **Smart contract deployment.** The TA deploys the smart contracts authentication contract and transaction contract on the blockchain. The addresses of the two smart contracts are published and those with the appropriate permissions can invoke the corresponding functions.

B. Registration

The vehicle i registers with TA and obtains a signature η . The step is as follows:

- The vehicle i runs $\Psi.PKIJoin(i, \lambda)$ (see Section IV-B) to obtain his public/private key pair (usk_i, upk_i) . The key pair is utilized to authenticate the messages sent by the vehicle to the TA.
- The vehicle i randomly chooses $sk_i \leftarrow \mathbb{Z}_p^*$, sets $(\gamma, \tilde{\gamma}) \leftarrow (g^{sk_i}, \tilde{Y}_1^{sk_i})$, computes $\sigma \leftarrow \Phi.Sign(\gamma, usk_i)$, computes the following zero-knowledge proof $ZKPOK_1$, which proves

that he knows the secret value sk_i . The vehicle sends $(\sigma, ZKPOK_1, \gamma, \tilde{\gamma})$ to the TA.

$$ZKPOK_1\{(sk_i) : \gamma = g^{sk_i} \wedge \tilde{\gamma} = \tilde{Y}_1^{sk_i}\}.$$

- The TA checks the validity of σ , $ZKPOK_1$ and checks whether $e(\gamma, \tilde{Y}_1) = e(g, \tilde{\gamma})$. If valid, the TA randomly chooses a unique identification $uid_i \in \mathbb{Z}_p^*$ for the vehicle. The TA computes $\eta \leftarrow (\eta_1, \eta_2) \leftarrow (g^t, (g^x \cdot \gamma^{y_1} \cdot g^{uid_i \cdot y_2})^t)$. Finally, the TA stores the entry $(i, \gamma, \tilde{\gamma}, \sigma, uid_i)$ in his own register, and sends (η, uid_i) to the vehicle. The vehicle i sets his private group key gsk_i as (sk_i, η) .
- The TA grants access to the vehicle i . Suppose there are ℓ authorized entries in the uselist, the TA first invokes $getAcc$ algorithm to get the latest accumulated value acc_ℓ . The TA computes $acc_{\ell+1} = acc_\ell^{ask+uid_i}$ and sets the vehicle's witness $w_{\ell+1} = acc_\ell$.
- The TA invokes $updateUser(uid_i, 1, acc_{\ell+1})$ algorithm to add the vehicle's unique identification in the userlist. When other vehicles already registered see the contract invocation on the blockchain, they update their own witness, i.e., a vehicle k with a witness $\hat{w}$ and his unique identification uid_k sets his new witness $\hat{w}^{new} = acc_\ell \cdot \hat{w}^{(uid_{\ell+1}-uid_k)}$.

C. Message Signing

In this stage, the vehicle produces a message/signature pair along with a encrypted pseudonym. The vehicle has the private group key $gsk_i = (sk_i, \eta)$ and the witness w . He also invokes $getAcc$ algorithm to get the latest accumulated value acc .

- The vehicle i first randomly chooses $r \in \mathbb{Z}_p^*$ and computes a re-randomized signature $(\eta'_1, \eta'_2) \leftarrow (\eta_1^r, \eta_2^r)$ which is actually a signature of sk_i .
- The vehicle selects a scope scp for this communication message m , randomly chooses $\alpha \in \mathbb{Z}_p^*$, and computes a encrypted pseudonym $cnym \leftarrow (cnym_1, cnym_2) \leftarrow (g^\alpha, dpk^\alpha \cdot \mathcal{H}_1(scp)^{sk_i})$.
- The vehicle computes a signature of proof SOK_1 (see Fig. 4) as a group signature π on the message m , and sends $(m, scp, \pi, cnym, time)$ to nearby vehicles or RSUs, where $time$ denotes the current time.

$$\begin{aligned} &SOK_1\{(sk_i, uid_i, w) : e(\eta'_1, \tilde{X} \cdot \tilde{Y}_1^{sk_i} \cdot \tilde{Y}_2^{uid_i}) = e(\eta'_2, \tilde{g}) \\ &\quad \wedge e(w, apk \cdot \tilde{h}^{uid_i}) = e(acc, \tilde{h}). \wedge cnym_1 = g^\alpha \wedge \\ &\quad cnym_2 = dpk^\alpha \cdot \mathcal{H}_1(scp)^{sk_i}\}(m||time). \end{aligned}$$

D. Message Verification

In this stage, given the tuple $(m, scp, \pi, cnym)$, the verifier (a vehicle or an RSU) can check the validity of π . If π is determined to be valid, the verifier accepts the message and uploads the data authentication information for further processing by the DP.

- The verifier checks the validity of π . If valid, the verifier accepts the traffic-related information m .
- The verifier parses $cnym$ as $(cnym_1, cnym_2)$, computes a transaction id $txid = \mathcal{H}_2(m)$ and invokes $addTx(txid, \pi, cnym_1, cnym_2)$ algorithm. The verifier also sends the message m to the DP.

$$\begin{aligned} &SOK_1\{(sk_i, uid_i, w) : e(\eta'_1, \tilde{X} \cdot \tilde{Y}_1^{sk_i} \cdot \tilde{Y}_2^{uid_i}) = \\ &\quad e(\eta'_2, \tilde{g}) \wedge e(w, apk \cdot \tilde{h}^{uid_i}) = e(acc, \tilde{h}) \wedge cnym_1 = \\ &\quad g^\alpha \wedge cnym_2 = dpk^\alpha \cdot \mathcal{H}_1(scp)^{sk_i}\}(m||time). \end{aligned}$$

Public information: $(\eta'_1, \eta'_2, \tilde{X}, \tilde{Y}_1, \tilde{Y}_2, \tilde{g}, \tilde{h}, apk, acc, cnym_1, cnym_2, scp)$.

Private information (witness): (sk_i, uid_i, w)

Let $T_1 = e(\eta'_1, \tilde{X})$, $T_2 = e(\eta'_1, \tilde{Y}_1)$, $T_3 = e(\eta'_1, \tilde{Y}_2)$, $T_4 = e(\eta'_2, \tilde{g})$, $T_5 = \mathcal{H}_1(scp)$.

Prover ($\mathcal{P}$):

1. $\mathcal{P}$ randomly chooses $\zeta_1, \zeta_2 \in \mathbb{Z}_p^*$, sets $\theta_1 = \zeta_1 \cdot uid_i$, $\theta_2 = \zeta_2 \cdot uid_i$, and computes $B_1 = g^{\zeta_1} h^{\zeta_2}$, $B_2 = w \cdot h^{\zeta_2}$.
2. Let $T_6 = e(B_2, apk)/e(acc, \tilde{h})$, $T_7 = e(h, \tilde{h})$, $T_8 = e(h, apk)$, $T_9 = e(B_2, \tilde{h})$.

3. It can be converted into $SOK_1\{(\zeta_1, \zeta_2, \theta_1, \theta_2, sk_i, uid_i, w, \alpha) : B_1 = g^{\zeta_1} h^{\zeta_2} \wedge B_1^{uid_i} = g^{\theta_1} h^{\theta_2} \wedge T_4/T_1 = T_2^{sk_i} T_3^{uid_i} \wedge T_6 = T_7^{\theta_2} T_8^{\zeta_2} T_9^{-uid_i} \wedge cnym_1 = g^\alpha \wedge cnym_2 = dpk^\alpha \cdot T_6^{sk_i}\}(m||time)$.

4. $\mathcal{P}$ randomly chooses $\rho_{\zeta_1}, \rho_{\zeta_2}, \rho_{\theta_1}, \rho_{\theta_2}, \rho_{sk_i}, \rho_{uid_i}, \rho_w, \rho_\alpha \in \mathbb{Z}_p^*$, and computes $D_1 = g^{\rho_{\zeta_1}} h^{\rho_{\zeta_2}}$, $D_2 = B_1^{-\rho_{uid_i}} g^{\rho_{\theta_1}} h^{\rho_{\theta_2}}$, $D_3 = T_2^{\rho_{sk_i}} T_3^{\rho_{uid_i}}$, $D_4 = T_7^{\rho_{\theta_2}} T_8^{\rho_{\zeta_2}} T_9^{\rho_{-uid_i}}$, $D_5 = g^{\rho_\alpha}$, $D_6 = dpk^{\rho_\alpha} T_6^{\rho_{sk_i}}$.

5. $\mathcal{P}$ computes the challenge $c = \mathcal{H}_2(D_1, D_2, D_3, D_4, D_5, D_6, m||time)$.

5. $\mathcal{P}$ computes the response $z_{\zeta_1} = \rho_{\zeta_1} - \zeta_1 \cdot c$, $z_{\zeta_2} = \rho_{\zeta_2} - \zeta_2 \cdot c$, $z_{\theta_1} = \rho_{\theta_1} - \theta_1 \cdot c$, $z_{\theta_2} = \rho_{\theta_2} - \theta_2 \cdot c$, $z_{sk_i} = \rho_{sk_i} - sk_i \cdot c$, $z_{uid_i} = \rho_{uid_i} - uid_i \cdot c$, $z_w = \rho_w - w \cdot c$, $z_\alpha = \rho_\alpha - \alpha \cdot c$ and sends the proof $\pi = (c, z_{\zeta_1}, z_{\zeta_2}, z_{\theta_1}, z_{\theta_2}, z_{sk_i}, z_{uid_i}, z_w, z_\alpha)$ to the $\mathcal{V}$.

Verifier ($\mathcal{V}$):

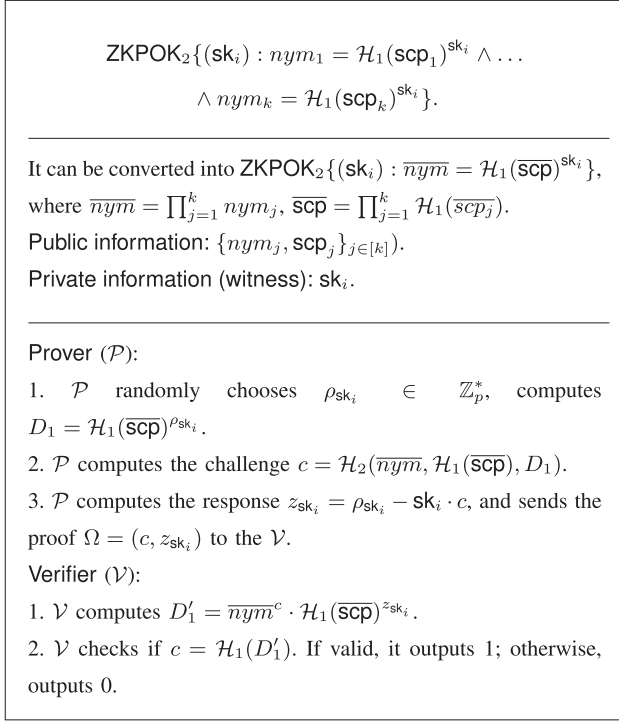
1. $\mathcal{V}$ computes $D'_1 = B_1^c g^{z_{\zeta_1}} h^{z_{\zeta_2}}$, $D'_2 = B_1^{-z_{uid_i}} g^{z_{\theta_1}} h^{z_{\theta_2}}$, $D'_3 = (T_4/T_1)^c T_2^{z_{sk_i}} T_3^{z_{uid_i}}$, $D'_4 = T_6^c T_7^{z_{\theta_2}} T_8^{z_{\zeta_2}} T_9^{-z_{uid_i}}$, $D'_5 = (cnym_1)^c g^{z_\alpha}$, $D'_6 = (cnym_2)^c dpk^{z_\alpha} T_6^{z_{sk_i}}$.
2. $\mathcal{V}$ checks if $c = \mathcal{H}_1(D'_1, D'_2, D'_3, D'_4, D'_5, D'_6, m||time)$. If valid, it outputs 1; otherwise, outputs 0.

Fig. 4. Construction of SOK_1 .

E. Message Link

In this phase, the DP can obtain *implicit* and *explicit* linkability of messages. DP received different messages $\{m_i\}_{i \in [n]}$ from vehicles or RSUs.

- For each message $\{m_i\}_{i \in [n]}$, the DP computes a transaction id $txid_i = \mathcal{H}_2(m_i)$, and invokes $getTx(txid_i)$ algorithm

Fig. 5. Construction of ZKPOK_2 .

to obtain $(g_{s_i}, \text{cnym}_{1,i}, \text{cnmy}_{2,i})$. Then, the DP sets g_{s_i} as the group signature π_i and $(\text{cnym}_{1,i}, \text{cnmy}_{2,i})$ as the encrypted pseudonym cnym_i .

- For $i \in [n]$, the DP computes the pseudonym $\text{nym}_i = \text{cnmy}_{2,i} / \text{cnmy}_{1,i}^{\text{dpk}}$, which implies $\text{nym}_i = \text{cnmy}_{2,i} / \text{cnmy}_{1,i}^{\text{dpk}} = \text{dpk}^\alpha \cdot \mathcal{H}_1(\text{scp})^{\text{sk}_i} / g^{\alpha \cdot \text{dsk}} = \mathcal{H}_1(\text{scp})^{\text{sk}_i}$. If $\text{nym}_i = \text{nym}_j, \forall i, j \in [n]$, it indicates that these two messages belong to the same vehicle for the same scope. The DP can analyze the information of the same vehicle and provide corresponding instructions to the vehicle or nearby RSUs (if needed).
- For explicit linkability, the vehicle i must provide an explicit link proof ZKPOK_2 for a list of his messages (Assuming there are k messages). The DP can check the validity of ZKPOK_2 (see Fig. 5). If valid, The DP can do the analysis work and provide instructions.

$$\text{ZKPOK}_2\{(\text{sk}_i) : \text{nym}_1 = \mathcal{H}_1(\text{scp}_1)^{\text{sk}_i} \wedge \dots$$

$$\wedge \text{nym}_k = \mathcal{H}_1(\text{scp}_k)^{\text{sk}_i}\}.$$

F. Tracking and Revocation

In this stage, the TA is able to open the signature and extract the identity i of the vehicle.

- The TA searches all entries $(i, \gamma_i, \tilde{\gamma}_i, \sigma_i, \text{uid}_i)$ in his own register whether $e(\eta_2, \tilde{g}) \cdot e(\eta_1, \tilde{X})^{-1} = e(\eta_1, \tilde{\gamma})$ holds. Until he gets a match, the TA knows the corresponding entry $(i, \gamma_i, \sigma_i, \text{uid}_i)$ where i is the vehicle's identity.

The vehicle's identity can be revoked as follows.

- The TA first invokes `getAcc` algorithm to get the latest accumulated value acc_ℓ . The TA computes $\text{acc}_{\ell+1} =$

$\text{acc}_\ell^{1/(\text{ask}+\text{uid}_i)}$. The TA invokes `updateUser`($\text{uid}_i, 0, \text{acc}_{\ell+1}$) algorithm to revoke the vehicle's unique identification in the userlist.

- When other vehicles already registered see the contract invocation on the blockchain, they update their own witness, i.e., a vehicle k with a witness $\hat{w}$ and his unique identification uid_k sets his new witness $\hat{w}^{\text{new}} = (\hat{w} / \text{acc}_{\ell+1})^{1/(\text{uid}_k - \text{uid}_i)}$.

VI. SECURITY ANALYSIS

In this section, we show that our scheme satisfies previously defined security requirements. As BAP mainly uses the ZKPOK and SOK to protect privacy. First, we prove that they satisfies the security properties of the NIZK argument, including completeness, soundness, and zero-knowledge. Second, we analyze the security requirements.

Theorem 1: The ZKPOK_1 and ZKPOK_2 constructed from Σ -protocol and Fiat-Shamir heuristic transformation satisfies the properties of completeness, soundness and zero-knowledge.

PROOF (SKETCH): These two ZKPOK s are used to prove simple discrete logarithmic relations, and since they are derived from the Σ -protocol via Fiat-Shamir heuristic transformation, the scheme is easy to prove the completeness, soundness and zero-knowledge, and we ignore it here.

Theorem 2: The SOK_1 constructed from Σ -protocol and Fiat-Shamir heuristic theorem satisfies the properties of completeness, soundness, zero-knowledge.

PROOF (SKETCH). We discuss completeness, soundness and zero-knowledge respectively.

Completeness: First, we need to prove the correctness of the converted SOK_1 . The converted signature of knowledge is $\text{SOK}_1\{(\zeta_1, \zeta_2, \theta_1, \theta_2, \text{sk}_i, \text{uid}_i, w, \alpha) : B_1 = g^{\zeta_1} h^{\zeta_2} \wedge B_1^{\text{uid}_i} = g^{\theta_1} h^{\theta_2} \wedge T_4/T_1 = T_2^{\text{sk}_i} T_3^{\text{uid}_i} \wedge T_6 = T_7^{\theta_2} T_8^{\zeta_2} T_9^{-\text{uid}_i}) \wedge \text{cnym}_1 = g^\alpha \wedge \text{cnym}_2 = \text{dpk}^\alpha \cdot T_6^{\text{sk}_i}\}$. Note $B_1 = g^{\zeta_1} h^{\zeta_2} \wedge B_1^{\text{uid}_i} = g^{\theta_1} h^{\theta_2}$, it implies $\theta_1 = \zeta_1 \cdot \text{uid}_i$, $\theta_2 = \zeta_2 \cdot \text{uid}_i$. Then, $T_4/T_1 = T_2^{\text{sk}_i} T_3^{\text{uid}_i} \implies e(\eta'_2, \tilde{g})/e(\eta'_1, \tilde{X}) = e(\eta'_1, \tilde{Y}_1)^{\text{sk}_i} e(\eta'_1, \tilde{Y}_2)^{\text{uid}_i} \implies e(\eta'_1, \tilde{X} \cdot \tilde{Y}_1^{\text{sk}_i} \cdot \tilde{Y}_2^{\text{uid}_i}) = e(\eta'_2, \tilde{g})$. Meanwhile,

$$\begin{aligned} T_6 &= T_7^{\theta_2} T_8^{\zeta_2} T_9^{-\text{uid}_i} \\ &\implies e(B_2, \text{apk})/e(\text{acc}, \tilde{h}) = e(h, \tilde{h})^{\theta_2} \\ &\quad \times e(h, \text{apk})^{\zeta_2} e(B_2, \tilde{h})^{-\text{uid}_i} \\ &\implies e(w \cdot h^{\zeta_2}, \text{apk} \cdot \tilde{h}^{\text{uid}_i})/e(\text{acc}, \tilde{h}) \\ &= e(h, \tilde{h})^{\zeta_2 \cdot \text{uid}_i} e(h, \text{apk})^{\zeta_2} \\ &\implies e(w, \text{apk} \cdot \tilde{h}^{\text{uid}_i}) e(h^{\zeta_2}, \text{apk} \cdot \tilde{h}^{\text{uid}_i}) \\ &= e(\text{acc}, \tilde{h}) e(h, \tilde{h})^{\zeta_2 \cdot \text{uid}_i} e(h, \text{apk})^{\zeta_2} \\ &\implies e(w, \text{apk} \cdot \tilde{h}^{\text{uid}_i}) = e(\text{acc}, \tilde{h}) \end{aligned}$$

Then, we have that the SOK_1 s before and after conversion are equivalent. Second, we need to prove that the process on the

converted SOK is correct. Since the SOK is used to prove simple discrete logarithmic relations, it is easy to prove its correctness.

Soundness: Assuming the discrete logarithm assumption, we can prove that the proposed scheme satisfies soundness under the random oracle model. Assuming a PPT prover $\mathcal{P}$ output an accepted transcript $\pi = (c, z_{\zeta_1}, z_{\zeta_2}, z_{\theta_1}, z_{\theta_2}, z_{sk_i}, z_{uid_i}, z_w, z_\alpha)$ to the $\mathcal{V}$. There exists a knowledge extractor $\mathcal{E}$. $\mathcal{E}$ can use rewinding to get another transcript $(c', z'_{\zeta_1}, z'_{\zeta_2}, z'_{\theta_1}, z'_{\theta_2}, z'_{sk_i}, z'_{uid_i}, z'_w, z'_\alpha)$. Further, the $\mathcal{E}$ can extract the witness:

$$\begin{aligned} \zeta_1 &= \frac{z_{\zeta_1} - z'_{\zeta_1}}{c' - c}, \quad \zeta_2 = \frac{z_{\zeta_2} - z'_{\zeta_2}}{c' - c}, \quad \theta_1 = \frac{z_{\theta_1} - z'_{\theta_1}}{c' - c}, \\ \theta_2 &= \frac{z_{\theta_2} - z'_{\theta_2}}{c' - c}, \quad sk_i = \frac{z_{sk_i} - z'_{sk_i}}{c' - c}, \\ uid_i &= \frac{z_{uid_i} - z'_{uid_i}}{c' - c}, \quad w = \frac{z_w - z'_w}{c' - c}. \end{aligned}$$

The adversary can use this algorithm to break the ECDLP with non-negligible probability. Thus, SOK_1 satisfies soundness.

Zero-knowledge: For the Fiat-Shmír heuristic transformation, we prove that there exists a simulator $\mathcal{S}$ which can simulate the interaction with a prover. $\mathcal{S}$ randomly chooses $c^*, z_{\zeta_1}^*, z_{\zeta_2}^*, z_{\theta_1}^*, z_{\theta_2}^*, z_{sk_i}^*, z_{uid_i}^*, z_w^*, z_\alpha^* \in \mathbb{Z}_p^*$, and computes

$$\begin{aligned} D_1^* &= B_1^{c^*} g^{z_{\zeta_1}^*} h^{z_{\zeta_2}^*}, \quad D_2^* = B_1^{-z_{uid_i}^*} g^{z_{\theta_1}^*} h^{z_{\theta_2}^*}, \\ D_3^* &= (T_4/T_1)^{c^*} T_2^{z_{sk_i}^*} T_3^{z_{uid_i}^*}, \quad D_4^* = T_6^{c^*} T_7^{z_{\theta_2}^*} T_8^{z_{\zeta_2}^*} T_9^{-z_{uid_i}^*}, \\ D_5^* &= (cnym_1)^{c^*} g^{z_\alpha^*}, \quad D_6^* = (cnym_2)^{c^*} dpk^{z_\alpha^*} T_6^{z_{sk_i}^*}. \end{aligned}$$

$\mathcal{S}$ views the hash function $\mathcal{H}_1$ as a random oracle and set the outputs $\mathcal{H}_1(c^*, z_{\zeta_1}^*, z_{\zeta_2}^*, z_{\theta_1}^*, z_{\theta_2}^*, z_{sk_i}^*, z_{uid_i}^*, z_w^*, z_\alpha^*, m || \text{time})$ to c^* . Since all numbers are selected at random, the transcript $(m, D_1^*, D_2^*, D_3^*, D_4^*, D_5^*, D_6^*, c^*, z_{\zeta_1}^*, z_{\zeta_2}^*, z_{\theta_1}^*, z_{\theta_2}^*, z_{sk_i}^*, z_{uid_i}^*, z_w^*, z_\alpha^*)$ are distributed identically with the a real one, which concludes the zero-knowledge.

Then, we analyze the properties of our proposed scheme.

- **Message authentication.** The EUF-CMA of PS signature ensures that the user's private group key gsk_i is unforgeable. Further, due to the completeness and soundness of SOK_1 , there exists no PPT adversary who can forge a valid message/signature pair in the Message Signing phase. Therefore, the verifier can check the validity of the signature in the Message Verification phase.
- **Privacy Preservation.** Privacy preservation is a paramount concern in our paper, given the sensitivity of vehicular data and the potential threats in VANETs. To ensure a comprehensive analysis of privacy, we conduct a thorough analysis of our protocol from the perspective of each entity involved.
- For an honest and curious TA, our protocol employs ZKPOK₁ to safeguard the secrecy of each vehicle's secret key (sk_i). This cryptographic technique enables the TA to grant access to vehicle i without having access to sk_i . As a result, the TA remains unaware of individual vehicle identities, reinforcing privacy protection.

- To address potential threats from malicious verifiers during the Message Verification phase, we utilize SOK_1 . This ensures that each vehicle possesses a valid signature on η , effectively concealing the real identity of the vehicle. Even with the TA's involvement in the process, the actual vehicle identity remains hidden from the TA and any potential eavesdroppers or malicious entities.
- For an honest and curious Data Provider (DP), our protocol limits the information accessible to them. The DP can only obtain pseudonyms of vehicles and has restricted access to linking information for the same vehicle in the case of implicit linkability. In the case of explicit linkability, the DP is a passive recipient of unsolicited information from the vehicle.

Our protocol ensures that sensitive vehicular data remains secure and anonymous, mitigating the risk of privacy breaches.

- **Conditional Traceability.** Only the TA can open can open the vehicle's identity when there is malicious behavior occurring. In this article, this property is ensured by searching the entries $(i, \gamma_i, \tilde{\gamma}_i, \sigma_i, uid_i)$ in his own register whether $e(\eta_2, \tilde{g}) \cdot e(\eta_1, \tilde{X})^{-1} = e(\eta_1, \tilde{\gamma})$ holds. Therefore, our scheme can effectively provides conditional traceability.
- **Unlinkability.** In the Message Signing phase, the vehicle can produce a group signature π , which ensures that it has a valid re-randomized signature η' of η , along with an encrypted pseudonym $cnym \leftarrow (cnym_1, cnym_2)$. Since the use of PS group signature, the signature η' is re-randomized, and $cnym$ is encrypted pseudonym of the pseudonym nym by using a random number r , there is no no PPT adversary who can link two messages sent by the same vehicle based on previous communication history.
- **Resistance against other common attacks:** Our scheme can resist against commonly encountered attacks in VANETs, such as replay attacks, modification attacks, and impersonation attacks.
 - **Impersonation attack:** Once an adversary wants to impersonate a legal vehicle, it needs to forge a valid signature of the TA. However, the message authentication property makes it difficult for a PPT adversary.
 - **Modification attack:** Since the completeness and soundness of SOK , the adversary cannot forge a valid group signature. Moreover, any modification can be discovered in the verification phase.
 - **Man-in-the-middle attack:** Since the message authentication property is given previously, the adversary cannot forge a valid signature without knowing sk_i . Thus, our scheme can defend this attack.
 - **Replay attack:** At each authentication, a new SOK_1 is generated and embedded with a current timestamp time . By checking the timestamp, it can be ensured that the scheme can resist replay attacks.
 - **Key escrow problem:** The identity-based system always leaks the user's private key if the authority is corrupted. However, in our scheme, even the TA is corrupted, the secret key of a vehicle (i.e., sk_i) is still preserved.

- Birthday collision resilience: We make use of a hash function $\mathcal{H}_2$ to generate a transaction id in transaction contract. Since the collision-resistant of hash functions (e.g., SHA256), our scheme can defend this attack.
- Implicit linkability: For the same scope, the vehicle's pseudonym is the same. This property comes straightforward. After the decryption, the DP get the message's $nym_i = \mathcal{H}_1(\text{scp})^{sk_i}$. If $nym_i = nym_i$, this necessarily indicates that the messages corresponding to the two pseudonyms belong to the same topic of the same vehicle.
- Explicit linkability: This property is ensured by active declaration of the vehicle. If the ZKPOK_1 is well-formed, then we have $\text{ZKPOK}_2\{(sk_i) : nym_1 = \mathcal{H}_1(\text{scp}_1)^{sk_i} \wedge \dots \wedge nym_k = \mathcal{H}_1(\text{scp}_k)^{sk_i}\}$. It implies that the messages corresponding to these pseudonyms belong to the different topic of the same vehicle.

VII. PERFORMANCE EVALUATION

In this section, we divided the discussion into three parts: blockchain cost, authentication cost, and simulation and results. We test BAP in a simulated scenario of VANET. The experimental results showed the feasibility of our scheme.

A. Blockchain Cost

We make use of Ethereum for our blockchain test network (i.e., Rinkeby). More specifically, we use a Remix which supports compiler (pragma solidity 0.4.25), language (Solidity), EVM version (compiler default). In Ethereum smart contracts, each operation (such as transferring funds, storing data) requires a certain amount of gas, which is a pricing unit in Ethereum used to pay for executing smart contract operations. Note that the smart contract method marked as view does not need to modify the state on the blockchain, so it does not consume any gas. This is because the view method only reads data on the blockchain and does not modify it.

We wrote two smart contracts authentication contract and transaction contract. Then, we compile and depoly them on the test network. Based on the contracts, we calculated the gas required for each operation and the corresponding USD cost (gas requires consumption of ether, which can be purchased in USD). We know that 1 gas needs 32 GWei and 1 GWei = 10^{-9} Ether, 1 Ether $\approx$ 1,878 USD.¹

We use “deploy1” and “deploy2” to represent the deployment of smart contracts authentication contract and transaction contract. The gas cost for these two operations is slightly larger, respectively. Fortunately, they only need to be operated at contract initialization phase, and getAcc and getTX algorithms are gas-free. As shown in Figs. 8 and 9, the gas costs of updateUser and addTX algorithms are 154426 (0.004941632 USD) and 365576 (0.011698432 USD), respectively, which demonstrates the practicality of our scheme.

Scalability: Evaluating the scalability of VANETs is crucial as these networks involve numerous vehicles generating significant

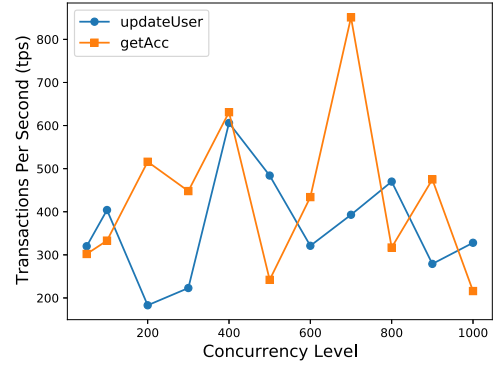


Fig. 6. TPS for authentication contract.

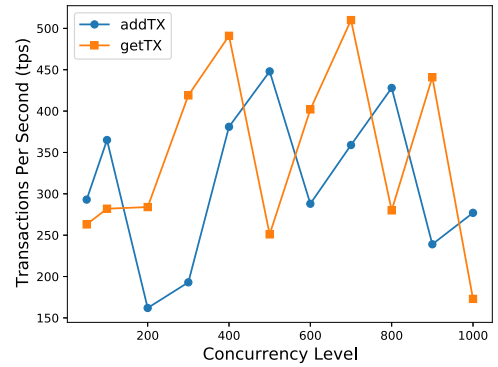


Fig. 7. TPS for transaction contract.

data. To achieve this, we utilize the Ethereum client, “geth” to deploy the smart contract representing our proposed protocol onto the blockchain network. Subsequently, we employ the CURL command-line tool to execute remote procedure calls (RPC) and interact with the deployed smart contract. To assess the scalability of our protocol, we conduct thorough load testing using the Apache HTTP server benchmarking tool, “ab”. We design a series of experiments with varying levels of concurrency (i.e., the number of simultaneous requests) to emulate real-world scenarios with increasing numbers of participants. By employing the “ab” tool with parameters -n 1000 (total number of requests) and -c 100 (concurrency level), we systematically test the protocol’s performance under different workloads.

As shown in the Figs. 6 and 7, the average TPS of the blockchain ranges from 350 to 450, making it suitable for accommodating the demands of VANETs applications. The results of our experiments demonstrate promising scalability characteristics, as the protocol maintains a relatively high TPS even with increased concurrency. This indicates that our proposed protocol can handle a substantial volume of transactions and participants while sustaining satisfactory performance. Indeed, while our solution has been locally tested using “geth,” acknowledging that real-world implementations may vary, our future research direction focuses on conducting practical evaluations in live vehicular ad hoc networks (VANETs).

Resource Requirements: Based on the operation of existing Ethereum clients [32], we list the minimum and recommended hardware specifications as follows:

¹<https://etherscan.io/gastracker>, accessed on April 26, 2023.

TABLE II
EXECUTION TIME AND DATA SIZE

Notation	Description	Time/Size
T_{pa1}	Time: Point addition in $\mathbb{G}_1$	0.00072 ms
T_{gm1}	Time: Point multiplication in $\mathbb{G}_1$	0.22 ms
T_{pa2}	Time: Point addition in $\mathbb{G}_2$	0.0023 ms
T_{gm2}	Time: Point multiplication in $\mathbb{G}_2$	0.37 ms
T_{pt}	Time: Pairing computation e	1.85 ms
T_{expt}	Time: Exponentiation in $\mathbb{G}_T$	0.95 ms
T_{h1}	Time: Hashing to G_1	0.21 ms
T_{h2}	Time: Hashing to $\mathbb{Z}_p^*$	0.002 ms
T_{abee}	Time: ABE encryption	390.87 ms
T_{abed}	Time: ABE decryption	77.02 ms
T_{zi}	Time: Modular Inversion in $\mathbb{Z}_p^*$	0.016 ms
T_{za}	Time: Modular addition in $\mathbb{Z}_p^*$	≈ 0 ms
T_{zm}	Time: Modular multiplication in $\mathbb{Z}_p^*$	≈ 0 ms
$ \mathbb{Z}_p $	Bit length of p	32 bytes
$ \mathbb{G}_1 $	Bit length of $\mathbb{G}_1$	32 bytes
$ \mathbb{G}_2 $	Bit length of $\mathbb{G}_2$	64 bytes
$ \mathbb{G}_T $	Bit length of $\mathbb{G}_T$	64 bytes
$ \text{time} $	Bit length of a timestamp	1 byte
$ \text{scp} $	Bit length of a scope	1 byte
$ \text{msg} $	Bit length of a message	-
$ \text{txid} $	Bit length of a transaction id	1 byte
$ \text{ABE} $	Bit length of an ABE ciphertext	$128\ell + 64$ bytes*

* ℓ is the number of leaf nodes in the access tree.

- Minimum Requirements: CPU: 2 cores or above, Memory: 8 GB, Available Hard Disk Space: 700 GB, Bandwidth: 10 MB/second or above.
- Recommended Hardware: CPU: 4 cores or above, with high speed, Memory: 16 GB or above, High-Speed Solid State Drive (SSD): 1 TB or above, Bandwidth: 25 MB/second or above.

B. Authentication Cost

In this section, we conduct our experiment on the laptop with an AMD Ryzen 7 6800H processor, clocking in at 3.20 GHz. It also comes equipped with 16 GB of RAM. The system operates on a 64-bit operating system, based on an x64 processor. We tested each common operation (e.g., point addition, point multiplication) by executing each operation 1000 times to take the average value, and the results are shown in Table II. we also summarized the size of the components in the scheme (also see Table II).

Further, we compare our scheme with the closest works BCPPA [4], BPAS [7] focused on providing privacy-preserving authentication protocol. For the computational cost, we compare our scheme with other schemes in detail in Table III and Fig. 10. Note that [7] provides additional access control, so the cost is larger. Both [7] and [4] cannot provide data linkability, which reduces the computational cost in this aspect. In our scheme, we use the precomputed techniques. Let $T_1 = e(\eta'_1, \tilde{X})$, $T_2 = e(\eta'_1, \tilde{Y}_1)$, $T_3 = e(\eta'_1, \tilde{Y}_2)$, $T_4 = e(\eta'_2, \tilde{g})$. Then, since $\eta'_1 = \eta_1^r$, the vehicle can first pre-compute $U_1 = e(\eta_1, \tilde{X})$, $U_2 = e(\eta_1, \tilde{Y}_1)$, $U_3 = e(\eta_1, \tilde{Y}_2)$, $U_4 = e(\eta_2, \tilde{g})$. When the vehicle

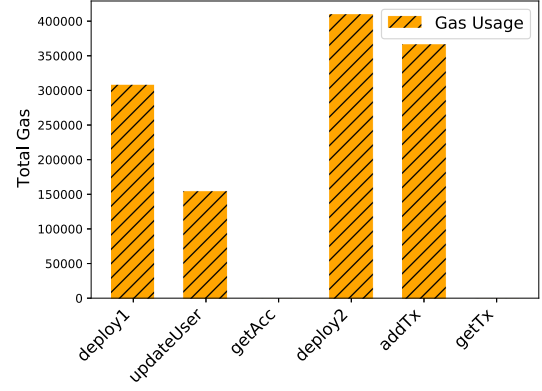


Fig. 8. Gas cost of our scheme.

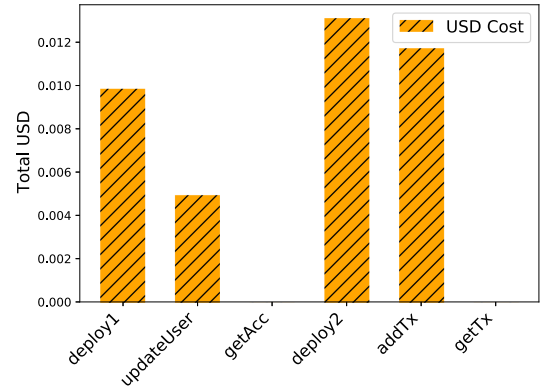


Fig. 9. USD cost of our scheme.

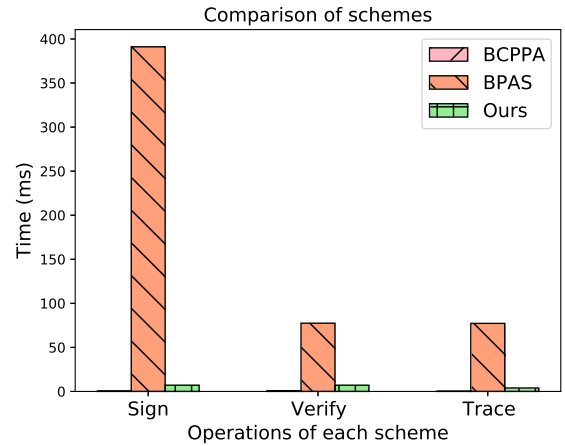


Fig. 10. Computational cost.

need to sign a message, he can directly calculate $T_1 = U_1^r$, $T_2 = U_2^r$, $T_3 = U_4^r$, $T_4 = U_4^r$. Note that $T_5 = \mathcal{H}_1(\text{scp})$, $T_7 = e(h, \tilde{h})$, $T_8 = e(h, \text{apk})$ also can be pre-computed. Since our solution supports additional features: implicit and explicit linkability, the computational cost of our scheme is in an acceptable range.

For the communication cost, we compare the schemes in Fig. 11. BCPPA needs to transmit a transaction id along with a signature and a timestamp. The total communication cost is $3|\text{txid}| + 4|\mathbb{Z}_p| + |\text{time}| + |\text{msg}|$ (264 bytes + $|\text{msg}|$). BPAS requires a ABE encryption and a signature to transferred. The

TABLE III
THEORETICAL TIME

Schemes	Sign	Verify	Trace
BCPPA [4]	$3T_{gm1}+T_{zi}+T_{zm}+T_{h2}$	$4T_{gm1}+T_{ga1}+T_{zm}+T_{h2}$	$k(T_{gm1} + T_{h2})^*$
BPAS [7]	$T_{abee}+T_{gm1}+T_{zm}+T_{h2}$	$T_{abed}+2T_{gm1}+T_{ga1}+T_{h2}$	$T_{abed}+T_{gm1}+T_{zm}$
Ours	$7T_{gm1}+17T_{ga1}+3T_{pt}$	$7T_{gm1}+17T_{ga1}+3T_{pt}$	$k(T_{gm1} + 2T_{pt})^*$

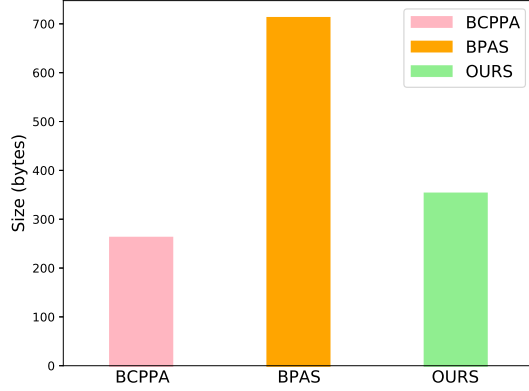
* k is the number of times to find entries.

Fig. 11. Signature size.

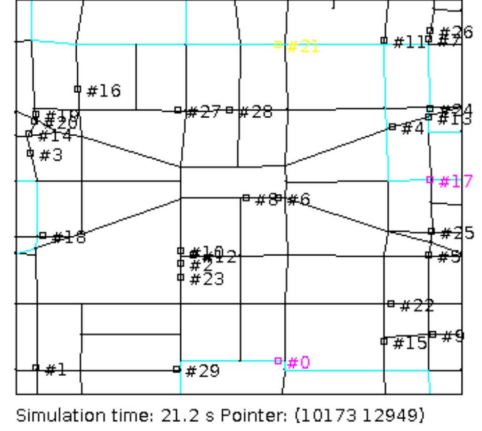


Fig. 13. Virtual map.

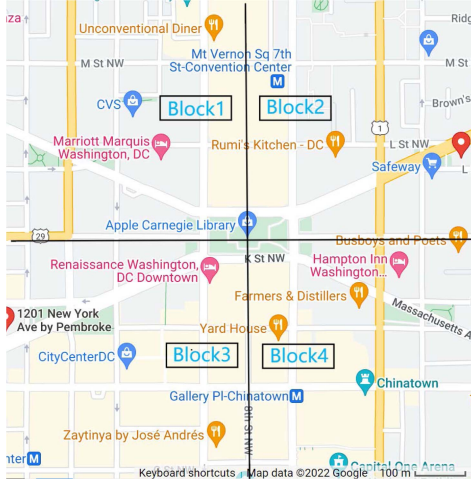


Fig. 12. Map of Washington, D.C. (USA).

total communication cost is $|G_1| + |Z_p| + |time| + |msg| + |ABE|$ (200 bytes + $128\ell + |msg|$), which is linear to ℓ . Our scheme BAP needs to send $(m, scp, \pi, cnym, time)$, whose size is $2|G_1| + 9|Z_p| + |time| + |msg| + |scp|$ (354 bytes + $|msg|$). Due to the need for additional encrypted pseudonym communication, the signature size of our scheme is between the 2 schemes.

C. Simulation and Results

In this section, we run simulations of our schemes. We use VanetMobiSim to simulate mobile nodes in VANETs as a way

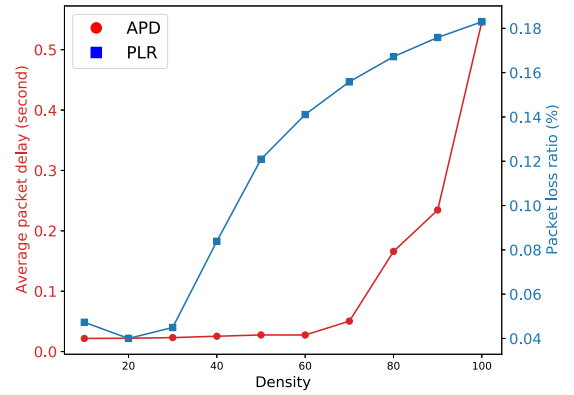


Fig. 14. APD and PLR with different densities.

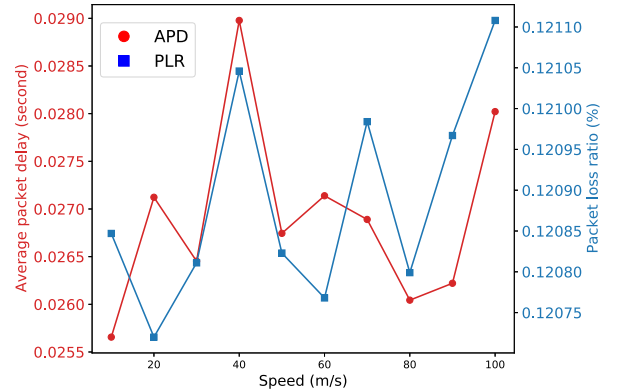


Fig. 15. APD and PLR with different speeds.

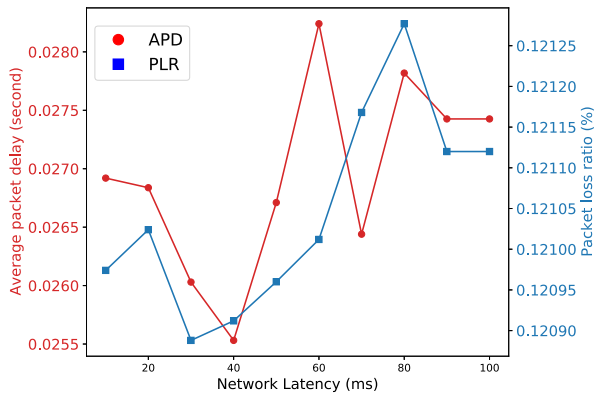


Fig. 16. APD and PLR with different network latency.

to evaluate the communication performance, routing protocols, and the effectiveness of communication protocols in vehicular networks. We use NS-2 to perform discrete-time simulations, which needs to set the number of vehicle nodes, the number of RSUs, and import map information. The network latency can be simulated by setting the transmission delay of the links during the creation of the network topology. Note that we use a $1.0 \times 1.0 \text{ km}^2$ map in Fig. 12 in Washington, D.C. (USA) whose Latitude and longitude are $(38.900015^\circ, -77.021515^\circ)$. We use a virtual map in Fig. 13 to perform the simulation.

We perform three types of simulations. First, we fix the speed of the vehicles at 30 m/s – 40 m/s and the transmission delay at 0 ms, then change the density of the vehicles from 5 to 100. We get the corresponding average packet delay (APD) and packet loss rate (PLR). As shown in Fig. 14, APD starts to rise at 30 vehicles, while PLR starts to rise rapidly at 60 vehicles. Second, we fix the number of the vehicles at 50 and the transmission delay at 0 ms, then change the speeds of the vehicles from 5 m/s to 100 m/s. As shown in Fig. 15, we find that APD and PLR vary dynamically and do not increase linearly with increasing vehicle speed. Third, we fix the number of the vehicles at 50 and the speed of the vehicles at 30 m/s – 40 m/s, then change the network latency from 10 m/s to 100 m/s. As shown in Fig. 16, The APD and PLR exhibit a positive correlation with the increase in network latency.

VIII. CONCLUSION

In this article, we introduced BAP, a privacy-preserving authentication protocol enhanced by blockchain technology. Our protocol utilizes PS group signature, ZKPOK, SOK, accumulator and blockchain techniques to ensure privacy protection. We employ accumulators for efficient registration and revocation of user identities. The use of blockchain enhances data security and mitigates the risk of data tampering by malicious participants. Through the security analysis, we verified the properties of ZKPOK and SOK protocols, as well as the overall security of BAP. The protocol supports message authentication, privacy preservation, traceability, and resistance against common attacks while providing user-controlled linkability. In the performance evaluation, we assessed blockchain, authentication, and

simulation costs. Overall, BAP demonstrated commendable efficiency and scalability.

Future work will focus on scalability, privacy-utility trade-offs, and real-world deployment to further advance the protocol's capabilities and applicability. We aim to address several aspects to further enhance BAP:

- **Scalability:** Exploring techniques like sharding and layer-two solutions to improve the protocol's scalability for large-scale deployments.
- **Privacy-Utility Trade-off:** Adjusting the protocol to strike a balance between privacy protection and utility remains a focus of future research.
- **Real-world Deployment:** Conducting real-world deployments and pilot studies in vehicular networks or other relevant scenarios will help validate BAP's practicality.

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